Practical – 5

Title :- Perform the Linear regression on the given data warehouse data using R

Theory :-

In Linear Regression these two variables are related through an equation, where exponent (power) of both these variables is 1. Mathematically a linear relationship represents a straight line when plotted as a graph. A non-linear relationship where the exponent of any variable is not equal to 1 creates a curve.

y = ax + b is an equation for linear regression.

Where, **y** is the response variable, **x** is the predictor variable and ***a*** and ***b*** are constants which are called the coefficients.

A simple example of regression is predicting weight of a person when his height is known. To do this we need to have the relationship between height and weight of a person.

The steps to create the relationship is −

- Carry out the experiment of gathering a sample of observed values of height and corresponding weight.
- Create a relationship model using the **lm()** functions in R.
- Find the coefficients from the model created and create the mathematical equation using these
- Get a summary of the relationship model to know the average error in prediction. Also called **residuals**.
- To predict the weight of new persons, use the **predict()** function in R.

Input Data

Below is the sample data representing the observations −

Values of height

151, 174, 138, 186, 128, 136, 179, 163, 152, 131

Values of weight.

63, 81, 56, 91, 47, 57, 76, 72, 62, 48

lm() Function

This function creates the relationship model between the predictor and the response variable.

Syntax

The basic syntax for **lm()** function in linear regression is −

lm(formula,data)

Following is the description of the parameters used −

- **formula** is a symbol presenting the relation between x and y.
- data is the vector on which the formula will be applied.

predict() Function

Syntax

The basic syntax for predict() in linear regression is −

predict(object, newdata)

Following is the description of the parameters used −

- **object** is the formula which is already created using the lm() function.
- newdata is the vector containing the new value for predictor variable.

Procedure

[1] **Create Relationship Model & get the Coefficients**

```
x <- c(151, 174, 138, 186, 128, 136, 179, 163, 152, 131)
```

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```
y <- c(63, 81, 56, 91, 47, 57, 76, 72, 62, 48)
# Apply the lm() function.
relation <- lm(y~x)
print(relation)
```

[2] **Get the Summary of the Relationship**

```
x <- c(151, 174, 138, 186, 128, 136, 179, 163, 152, 131)
y <- c(63, 81, 56, 91, 47, 57, 76, 72, 62, 48)
# Apply the lm() function.
relation <- lm(y~x)
print(summary(relation))
```

[3] **Predict the weight of new persons**

```
# The predictor vector.
x <- c(151, 174, 138, 186, 128, 136, 179, 163, 152, 131)
# The resposne vector.
y <- c(63, 81, 56, 91, 47, 57, 76, 72, 62, 48)
# Apply the lm() function.
relation <- lm(y~x)
# Find weight of a person with height 170.
a <- data.frame(x = 170)
result <- predict(relation,a)
print(result)
```

[4] **Visualize the Regression Graphically**

```
# Create the predictor and response variable.
x <- c(151, 174, 138, 186, 128, 136, 179, 163, 152, 131)
y <- c(63, 81, 56, 91, 47, 57, 76, 72, 62, 48)
relation <- lm(y~x)
# Give the chart file a name.
png(file = "linearregression.png")
# Plot the chart.
plot(y,x,col = "blue",main = "Height & Weight Regression",
abline(lm(x~y)),cex = 1.3,pch = 16,xlab = "Weight in Kg",ylab = "Height in cm")
# Save the file.
dev.off()
```

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Practical – 6

Title : Perform the logistic regression on the given data warehouse data using R.

Theory –

- Logistic Regression is a **supervised learning algorithm** used for **binary classification problems** (e.g., yes/no, success/failure, churn/no churn).
- Unlike linear regression, which predicts continuous values, logistic regression predicts the **probability of belonging to a class**.
- It is widely applied in business analytics, healthcare, marketing, and data warehouse environments where large-scale structured data is stored.

Logistic Regression in Data Warehousing

- Data warehouses store **large, integrated datasets** from multiple sources.
- Logistic regression is often applied to warehouse data for:
  - Customer churn prediction
  - Fraud detection
  - Purchase likelihood modeling
  - Risk assessment
- Preprocessing steps include:
  - Extracting relevant attributes (ETL process)
  - Handling missing values
  - Normalizing/transforming variables
  - Splitting into training and testing sets

Procedure :

[1] Importing the Dataset

```
install.packages("dplyr")
library(dplyr)

head(mtcars)
```

[2] Splitting the Dataset

We are using the caTools package to randomly split the mtcars dataset into two parts: 80% for training (train_reg) and 20% for testing (test_reg). This allows us to train the logistic regression model on one set and evaluate its performance on unseen data.

```
install.packages("caTools")
library(caTools)

split <- sample.split(mtcars, SplitRatio = 0.8)

train_reg <- subset(mtcars, split == "TRUE")
test_reg <- subset(mtcars, split == "FALSE")
```

[3] Building the model

Logistic regression is implemented in R using glm() by training the model using features or variables in the dataset.

```
logistic_model <- glm(vs ~ wt + disp, data = train_reg, family = "binomial")
logistic_model

summary(logistic_model)
```

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